Applied Surface Science 614 (2023) 156161

Contents lists available at ScienceDirect

Applied Surface Science

journal homepage: www.elsevier.com/locate/apsusc

Full Length Article

The initial oxidation of the 4H-SiC (0001) surface with C-related point defects: Insight by first-principles calculations

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ARTICLE INFO

Keywords:
Initial oxidation
4H-SiC surface
First-principles
Thermodynamics
Defects evolution

ABSTRACT

The initial oxidation of 4H-SiC (0001) surfaces with C-related point defects ($Surf_{CD}$) is studied by using first-principles coupled with thermodynamics calculations to gain deeper insights into the effects of C-related point defects on the formation of SiO_2 and SiO_2/SiC interface. Three types of $Surf_{CD}$, including a surface with C interstitial defect ($Surf_{Ci}$), a surface with antisite defect ($Surf_{Csi}$) and a surface with C vacancy ($Surf_{Vc}$) are considered. The initial oxidation of $Surf_{CD}$ commences on the chemisorption of oxygen atoms at the bridge sites between the Si atoms adjacent to the C-related point defects. As the oxygen coverage increased, the effects of C-related point defects on the adsorption sites and energies of oxygen atoms, the diffusion of oxygen atoms and the interfacial lattice strain are elucidated. The thermodynamic diagrams of O/4H-SiC structures are performed to investigate the stability of surface during the actual oxidation conditions for $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$. Thermodynamically stable intermediate structures are observed on these surfaces during the initial formation of SiO_2, the 1 ML structures are the most stable; the Si dangling bonds, Si-O-C structure and C-clusters are observed in the 1 ML structures. Moreover, the evolution mechanisms and electronic properties of the defects are further investigated.

1. Introduction

Silicon carbide (SiC) has generated considerable attention in high-power and high-frequency electronic devices due to its superior physical characteristics, including large bandgap, high breakdown field and thermal conductivity [1–4]. Moreover, the stable SiO_2 formed by thermal oxidation allows the fabrication of SiC metal–oxidesemiconductor field-effect-transistors (MOSFETs) using the conventional Si-based technology [5,6]. However, the SiC MOSFETS have unacceptably bias temperature instability (BTI) and poor reliability caused by high SiO_2/SiC interface trap density (D_{it}) [7–11]. The experimental results have indicated that thermal oxidation is one of the main reasons for forming interfacial defects [12–15], thus understanding the formation of SiO_2 on atomic scale is essential for deeper insight into interfacial defects and improving the properties of devices. Although previous studies have made valuable progress in studying the initial oxidation of SiC surface at submonolayer coverage of oxygen atoms [16,17], these works are mainly based on the ideal clean SiC surface, which reflects hardly the oxidation of true SiC surface accurately.

The morphologies and defects on the SiC surface are complicated, even after surface cleaning and treatments, there still contains terraces, lattice reconstruction and defects [18–20]. The surface conditions of SiC affect the formation of initial SiO_2 and SiO_2/SiC interfacial quality critically [21–23]. Recent theoretical studies showed that the vacancy and antisite defects on the 3C-SiC surface could introduce new C-containing gaseous oxide and covalent bonds during the initial oxidation [24]. Yamasaki et al. found that the numerous C–C defects generated below the SiO_2/SiC interface in the oxidation process of 4H-SiC (000_1) surface with π-bonded-chain structure [25]. Li et al. studied the initial oxidation of the stepped 4H-SiC surface and performed that the one-dimensional -Si-O- chain structure formed on the stepped surface was the precursor of the SiO_2 [26]. These works showed that initial defects or reconstruction on the SiC surface could partly account for the mechanism of poor SiO_2/SiC interfacial quality and formation of defects. Some work has also proved that the Si- and N-related defects on the surface have the significant effects on the adsorption of O atoms on the SiC surface [12,27]. However, atom-scale studies on the oxidation of defective 4H-SiC (0001) surfaces have rarely been performed, which is

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https://doi.org/10.1016/j.apsusc.2022.156161
Received 4 November 2022; Received in revised form 8 December 2022; Accepted 18 December 2022
Available online 20 December 2022
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indispensable for considering gate oxide growth in MOS devices [28,29]. In addition, the C-related defects on the 4H-SiC (0001) surface still existed after Radio Corporation of America (RCA) cleaning, which had been proven by XPS measurements [30,31]. C-related point defect is the typical defect of numerous C-related defects, which has the low formation energies and is considered an electroactive defect that affects device performance [32–35]. To clarify the effect of surface C-related point defects on the formation of SiO_2 and SiO_2/SiC interface, further systematic studies on the initial oxidation of the 4H-SiC (0001) surface with C-related point defects ($Surf_{CD}$) are required.

The main goal of this work is to investigate the absorption of O atoms on the 4H-SiC (0001) $Surf_{CD}$, the effects of C-related point defects during initial oxidation, the thermodynamic stability of O/4H-SiC structures and defect evolution of surface defects by density-functional theory (DFT) calculations. We performed this work by adding O atoms on the 4H-SiC (0001) $Surf_{CD}$ with coverage of O atoms from 1/9 to 1 monolayer (ML). Three types of $Surf_{CD}$, including a surface with C interstitial defect ($Surf_{Ci}$), a surface with antisite defect ($Surf_{Csi}$) and a surface with C vacancy ($Surf_{Vc}$) were considered. We studied the plausible adsorption site and adsorption energy of single O atom on these $Surf_{CD}$, investigated the adsorption behavior of O atoms on $Surf_{CD}$ with different O coverages during the initial oxidation. Moreover, the variations of structures as the oxidation process of $Surf_{CD}$ continues were discussed, and the effects of C-related defects on the diffusion of O atoms and the interfacial lattice strain were elucidated. We also examined the thermodynamic stability of the O/4H-SiC structure at actual oxidation conditions (oxidation temperature and pressure). Further, we studied the origin and electronic properties of defects generated in the initial oxidation, such as the Si dangling bonds, Si-O-C structure and C-clusters.

2. Computational method and models

All the DFT calculations in this work were performed using Vienna *ab initio* simulation package (VASP) [36]. The Perdew-Burke-Ernzerhof functional within the generalized gradient approximation (GGA-PBE) was used to describe the exchange correlation interactions [37]. A Gamma k-point mesh (separation was 0.04 $Å^{-1}$) and a kinetic-energy cutoff of 500 eV were adopted in all calculations. The convergence energy and force were 10^{-5} eV and 0.01 eV/Å for geometry optimization.

To study the effect of C-related surface point defects on the initial oxidation of 4H-SiC (0001) surface, we selected three different $Surf_{CD}$ models, including the $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$. The top views of optimized configurations for these three surfaces are depicted in Fig. 1(b), (c) and (d), respectively. The local defective configurations of $Surf_{Ci}$, $Surf_{Csi}$, and $Surf_{Vc}$ are shown in the right side of Fig. 1(b)–(d). In our previous work, we have demonstrated that these three defects were the most stable and have the lowest formation energy among the same types of defects [35]. All the slab models of 4H-SiC (0001) surface contained six Si-C bilayers, a hydrogen atomic layer at the bottom and a ~15 Å vacuum space (Fig. 1(a)). It has been proved that the hydrogen atoms and vacuum space set-up is enough to exclude the influence of bottom dangling bonds and periodic along vertical direction [38,39]. During the geometry optimization, only the upper two Si-C bilayers remained free to relaxation, the lower four bilayers and hydrogen atoms were fixed.

The initial oxidation of 4H-SiC begins with the adsorption of O atoms on the surface [16,27]. The adsorption of single O atom and multiple O atoms on the $Surf_{CD}$ were studied, the number of O atoms from 1 to 9 corresponded to the variation of O coverage from 1/9 to 1 ML. Considering that there was a substantial amount of Si dangling bonds on the initial surface, the formation and emission of CO or CO_2 were neglected in the calculation of oxygen atom adsorption for the O coverage below 2/3 ML. Nevertheless, there were still many adsorption sites of O atoms on these $Surf_{CD}$, since C-related point defects changed the chemical environment around each Si atom and destroyed the symmetry of the adsorption sites on the surface. The adsorption energy (E_{ads} (O)) per O atom on the $Surf_{CD}$ was calculated as follows:

$$E_{ads}(O) = \frac{1}{N_O}\left[E_{surf/O} - E_{surf} - \frac{N_O E_{tot}(O_2)}{2}\right] \quad (1)$$

where $E_{surf/O}$ and E_{surf} are the total energy of 4H-SiC surface model with and without O atoms adsorbed. N_O is the number of O atoms. E_{tot} (O_2) is the energy of a O_2 molecule, the calculated value of E_{tot} (O_2) is −9.847 eV.

Atomistic thermodynamics is an effective method to determine the surface stability at different temperature and pressure. This method has been widely used to study the stability of structures adsorbing atoms or molecules (H, O, H_2O, etc.) on the surface of binary or ternary materials [16,40–42]. Here, we considered the thermodynamics stability of $Surf_{CD}$ with different O coverages under temperature T and pressure P. The surface free energy $\gamma_{surf/O}$ was calculated by Eq. (2):

$$\gamma_{surf/O}(T,P) = \frac{1}{S}\left[G_{surf/O}(T,P) - N_{Si}\mu_{Si}(T,P) - N_C\mu_C(T,P) - N_O\mu_O(T,P)\right] \quad (2)$$

Where S is the surface area of the surface model. N_{Si} and N_C are the

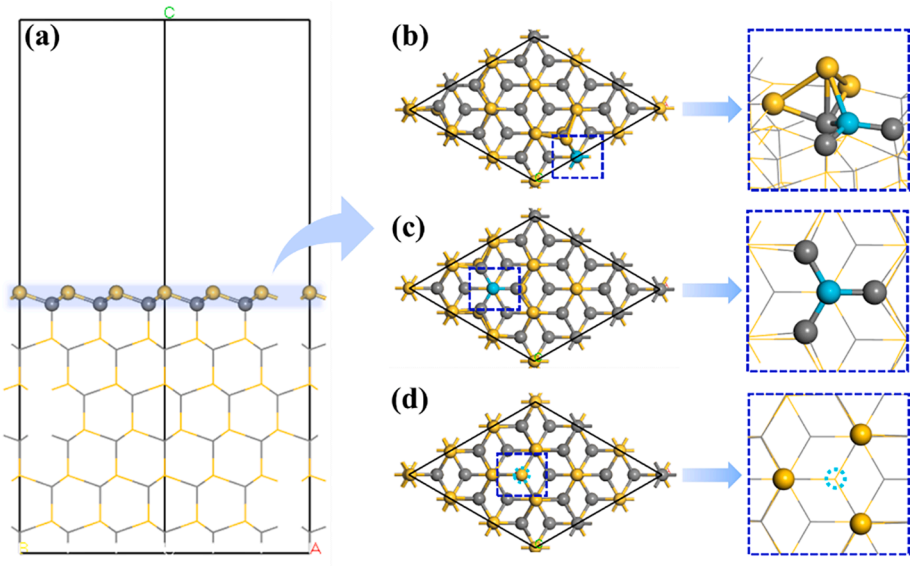


Fig. 1. (a) The side view of 4H-SiC (0001) surface slab, the area marked by the blue dashed box is the first Si-C bimolecular layer on the surface. (b) The top view of $Surf_{Ci}$. (c) The top view of $Surf_{Csi}$. (d) The top view of $Surf_{Vc}$. The local defective configurations of $Surf_{Ci}$, $Surf_{Csi}$, and $Surf_{Vc}$ are shown in the right side of Fig. 1(b)–(d), and marked by the bule dashed box. Si atom, C atom and C defective atom are represented by yellow ball, gray ball and indigo ball, respectively. The indigo dashed circle indicates the vacancy of C atom. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

number of Si and C atoms. μ_{Si}, μ_C and μ_O are the chemical potentials of Si, C and O atoms, respectively. $G_{surf/O}$ is the Gibbs free energy, which is theoretically contributed by $E_{surf/O}$, vibrational energy (F^{vib}), energy of volume change (PV) and configurational entropy (TS^{conf}). Due to the F^{vib}, PV and TS^{conf} are very low for solid materials, the contributions of these three terms can be neglected [43–46], namely, $G_{surf/O} \approx E_{surf/O}$. Therefore, the Eq. (2) could be rewritten as:

$$\gamma_{surf/O}(T,P) = \frac{1}{S}\left[E_{surf/O} - N_{Si}\mu_{Si}(T,P) - N_C\mu_C(T,P) - N_O\mu_O(T,P)\right] \quad (3)$$

The surface free energy of the $Surf_{CD}$ after O adsorption relative to that before O adsorption was defined as:

$$\begin{aligned}\Delta\gamma(T,P) &= \gamma_{surf/O}(T,P) - \gamma_{surf}(T,P) \\ &= \frac{1}{S}\left[E_{surf/O} - E_{surf} - \Delta N_{Si}\mu_{Si}(T,P) - \Delta N_C\mu_C(T,P) - \Delta N_O\mu_O(T,P)\right]\end{aligned} \quad (4)$$

Here ΔN_C, ΔN_{Si} and ΔN_O are the difference in the number of C, Si and O atoms on the O/4H-SiC surface relative to 4H-SiC surface. μ_{Si} and μ_C are linked by $G_{SiC} = \mu_{Si} + \mu_C$, the G_{SiC} is the Gibbs free energy of a bulk 4H-SiC unit. In Eq. (4), the parameter most closely related to temperature and pressure is μ_O (T, P). We treated O_2 as an ideal gas, then μ_O (T, P) was calculated by:

$$\mu_O(T,P) = \frac{1}{2}\left[E_{tot}(O_2) + \mu_{O_2}(T,P^0) + k_B T\ln\left(\frac{P}{P^0}\right)\right] \quad (5)$$

where P^0 refers to 1 atm, μ_{O2} (T, P^0) is the standard chemical potential of O_2, which can be obtained from the JANAF thermodynamic tables [47]. k_B is the Boltzmann constant. The μ_O (T, P) at $T = 0$ K was obtained by:

$$\mu_O(0K,P) = \frac{E_{tot}(O_2)}{2} = \mu_O^{MAX} \quad (6)$$

Which is the maximum value for chemical potential of O. Here, the lower boundary of μ_O was determined by Eq. (7) as follow:

$$\mu_O^{MIN} = \frac{E_{tot}(SiO_2) - E_{tot}(Si)}{2} \quad (7)$$

where E_{tot} (SiO_2) and E_{tot} (Si) are the energies of a bulk SiO_2 unit and a bulk Si unit, respectively. The range of μ_O obtained by DFT calculation was $-9.16 \le \mu_O \le -4.92$ eV. That is, SiO_2 was stable only when the chemical potential of O in this range.

In order to analyze the electronic properties of interface defects, the total density of states (TDOS), local density of states (LDOS) and partial charge density were calculated in this work. All the calculations of DOS were adopted Heyd-Scuseria-Ernzerhof (HSE06) hybrid functional to accurately estimate the bandgap [48]. 25% of PBE was replaced with Hartree-Fork function in our calculation. The calculated bandgap for 4H-SiC was 3.20 eV, which is approximately equal the experimental value of 3.26 eV.

3. Results and discussion

3.1. Adsorption of single O atom on the $Surf_{CD}$

We first analyzed the adsorption of O_2 molecule on the $Surf_{CD}$. The optimized configurations as shown in Supplementary Fig. S1, O_2 molecule spontaneously decomposed into two O atoms adsorbed on different sites on all $Surf_{CD}$. This result indicates that the configuration of O_2 molecule adsorbed on the $Surf_{CD}$ is unstable and relaxation eventually leads to the dissociation of the O_2 molecule, which is consistent with the finding that O_2 molecule adsorbed on the clean 4H-SiC surface [26,49].

Five different types of adsorption sites were considered to investigate the most stable adsorption sites of single O atom on the $Surf_{CD}$, including the top site of surface Si atom (T), bridge site between two Si atoms on the surface (B), fcc hollow site above the surface SiC hexagon (F), hcp hollow site above the surface SiC hexagon (H), other sites around C-related point defects (O). As shown in Fig. 2(a–c), these O adsorption sites on the $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$ were marked with dashed circles of different colors. The adsorption of the O atom on $Surf_{CD}$ is more complex than on clean surface because the C-related point defects decrease high symmetry adsorption sites of the O atom on the surface. For the adsorption of single O atom on the $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$, we considered 28, 14 and 15 possible configurations, respectively. After optimization, all the stable configurations are depicted in Supplementary Figs. S2, S3 and S4. Almost all F and H sites of the initial configurations were transformed to B sites, illustrating that the F and H adsorption sites of single O atom on the $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$ were unstable. It is inconsistent with the existence of four possible stable adsorption sites (T, B, F and H) of single O atom on the clean 4H-SiC (0001) surface [16,50], which may be related to the variation of atomic distribution and lattice strain on the local surface caused by C-related point defects. This is due to that, there is difference between the atomic arrangement of the surface with defects and that of the next substrate surface, which causes the strain in the near-surface region, further generating lateral and normal forces on the surface; while the adsorption of O atom on the surface will also induce local lateral and normal forces, the two forces on the surface interact with each other [51,52].

The distribution of E_{ads} of single O atom at different sites calculated according to Eq. (1) is shown in Fig. 3(a–c). In the same system, the lower the E_{ads} represents the stronger interaction between the adsorbed atoms and the surface; the lower the E_{ads}, the more stable the configuration is. For $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$, the lowest E_{ads} of a single O atom were −4.133, −4.228 and −4.781 eV, respectively. Compared with the clean 4H-SiC (0001) surface (−4.432 eV), the E_{ads} of single O atom on the $Surf_{Ci}$ and $Surf_{Csi}$ were higher, and that on the $Surf_{Vc}$ was lower, which indicates the existence of V_C is beneficial to the adsorption of single O atom on the 4H-SiC surface. In contrast, the adsorptions of O atom on the $Surf_{Ci}$ and $Surf_{Csi}$ were energetically less favorable compared with clean surface. The difference of C-related point defects on adsorption energy may cause different oxidation rates at the initial oxidation, further affecting the thickness uniformity of the oxide film. In addition, the most stable adsorption site of single O atom on these $Surf_{CD}$ was the B site (Fig. 3(d–f)), consistent with the type of the most stable adsorption site on the clean 4H-SiC surface [16]. The difference is that the most stable B site on $Surf_{CD}$ is located between the Si atoms connected to the C-related point defects.

3.2. Adsorptions of O atoms on the $Surf_{CD}$ with different coverages

As the oxidation process continued, more O atoms were introduced into the 4H-SiC surface. The increase of O coverages affected the configuration and energy of O adsorption on the surface, and the type and structure of surface defects were also changed. In this section, we studied the adsorption of O atoms on the $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$ with O coverage ranging from 1/9 to 1 ML. After sampling numerous configurations, we selected the lowest-energy configurations under different O coverage as shown in Figs. 4, 5 and 6, respectively.

Fig. 4(a–i) displays the lowest-energy O adsorption configurations on $Surf_{Ci}$ for O coverage from 1/9 to 1 ML. Our previous work has proved that the optimized C interstitial defect was a composite defect composed of C antisite (C_{Si}) and Si adatom, the Si adatom connected with C_{Si} and two Si atoms adjacent to C_{Si} through Si-C and Si-Si bonds, respectively [35]. As observed in Fig. 4(a) and (b), the most stable structure with the O coverage of 1/9 and 2/9 ML were the Si-O-Si structures formed by that the introduced O atoms destroyed the two Si-Si bonds. For the O coverage between 1/3 to 5/9, the most favorable absorption sites of O atoms were on the Si atoms with dangling bond (Fig. 4(c–e)). For the O coverage of 5/9 ML, all Si dangling bonds on the surface were saturated by O atoms, and the Si-C bond between the surface and subsurface SiC bilayer was broken, resulting in the generation of Si dangling bonds on the subsurface bilayer. With O coverage increasing up to 2/3 ML, the Si-

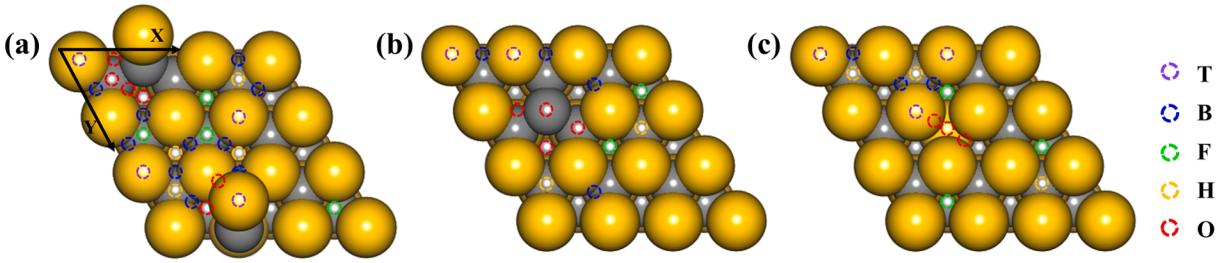


Fig. 2. Top views of the adsorption sites for single O atoms on (a) $Surf_{Ci}$, (b) $Surf_{Csi}$, (c) $Surf_{Vc}$. The purple, blue, green, yellow and red dotted circles represent the positions of adsorption sites T, B, F, H and O, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

C bond between C_{Si} and Si adatom was broken, the surface C interstitial defect was transformed into C_{Si} defect and O-Si-O structure absorbed on the surface, the introduced O atom was preferentially absorbed on the top site of the Si atom in the O-Si-O structure (Fig. 4(f)). When the O coverage reaches 7/9 ML, the SiO_4 tetrahedral structure with Si adatom as the center was obtained on the surface. The average Si-O bond length and O-Si-O bond angle in the SiO_4 tetrahedral were 1.67 Å and 116.24°, which are close to the structural parameters of SiO_2 measured in the experiment [53,54]. Thus, the SiO_4 structure may be the initial nucleation precursor of thermally oxidized SiO_2 on the $Surf_{Ci}$ (Fig. 4(g)). When the O coverage was increased to 8/9 or 1 ML, the introduced O atom tended to destroy the Si-C bond on the surface SiC bilayer, forming the Si-O-C structures on the surface (Fig. 4(h-i)). In addition, we considered the surface CO emission for O coverage above 5/9 ML. The results showed that the stable structure of CO emission existed only when the O coverage was 1 ML (Supplementary Fig. S5(a)), but the formation energy of this stable structure lay 1.197 eV above that of the structure shown in Fig. 4(i). We propose that the CO emission are energetically less favorable when the O coverage is below 1 ML on the $Surf_{Ci}$.

Fig. 5(a–i) displays the lowest-energy O adsorption configurations on $Surf_{Csi}$ for O coverage from 1/9 to 1 ML. The surface C_{Si} defect was formed by the C atom substituting the Si atom [35]. As depicted in Fig. 5 (a–c), the preferential adsorption sites of the introduced O atoms were the B sites between Si atoms adjacent to C_{Si} for the O coverage from 1/9 to 1/3 ML. At 4/9 ML, the Si dangling bonds on the surface were saturated by O atoms (Fig. 5(d)). When the O coverage reached 5/9 ML, the O atoms were prone to destroy the Si-C bonds on the surface, and the Si-O-C structure containing an unsaturated C atom was formed (Fig. 5(e)). As the O coverage increasing, more Si-C bonds were broken to generate more unsaturated C atoms, these unsaturated C atoms combined to form C-cluster defects, such as C_2-, C_5- and C_6-clusters depicted in Fig. 5(f–i). With O coverage increasing up to 1 ML, the O atom diffused to the subsurface, and the Si-O-Si structure was obtained between bilayers. Furthermore, the stable structures of CO emission existed at 2/3 ML and 7/9 ML O coverage (Supplementary Fig. S5(b) and (c)), and the corresponding formation energies of these stable structures were 3.76 eV and 2.21 eV higher than that of the structure shown in Fig. 5(f) and (g) respectively. This result indicates that CO emission is less likely to occur at the initial oxidation for $Surf_{Csi}$, the O atoms tend to absorb on the surface and diffuse to the subsurface.

Fig. 6(a–i) displays the lowest-energy O adsorption configurations on $Surf_{Vc}$ for O coverage from 1/9 to 1 ML. Due to the existence of C vacancy, $Surf_{Vc}$ had more Si dangling bonds than the clean 4H-SiC surface, resulting in the introduced O atoms were more preferentially bonded with the surface Si atoms, which is consistent with the lowest E_{ads} of single O atom on the $Surf_{Vc}$ calculated in section **3.1**. For O coverage of 1/9 ML, Fig. 6(a) and Supplementary Fig. S6(a) show that the O atom preferentially bands with the Si atoms around the C vacancy, and accompanied by the formation of Si-Si bond between bilayers. When the O coverage reached 1/3 ML, the Si-C bond between bilayers was broken (Fig. 6(c)). As shown in Supplementary Fig. S6(b), the stretching effect of O and Si atoms on Si1, Si2 and Si3 atoms around C1 atom caused the upward movement of C1 atom, breaking the Si-C between C1 and Si4 atoms (marked by dashed line). As the O coverage increased to 2/3 ML, most of the O atoms were bonded with the Si atoms around C vacancy (Fig. 6(f)), causing large local surface strain and forming the C–C defect as displayed in Supplementary Fig. S6(c). When the O coverage reached 7/9 ML, the Si-O-Si structure was formed by the diffusion of the introduced O atom to the subsurface, as shown in Fig. 6(g). All the Si dangling bonds were saturated at O coverage of 8/9 ML (Fig. 6(h)), which requires more O atoms than that of clean surface. When the O coverage reached 1 ML, the Si-C bonds were broken by O atoms to form Si-O-C structures (Fig. 6(i)). Moreover, the formation energy of the stable configuration of CO emission (Supplementary Fig. S5(d)) was 1.913 eV more than that of the structure depicted in Fig. 6(i). Analogous to $Surf_{Ci}$ and $Surf_{Csi}$, the emission of CO is almost impossible on the $Surf_{Vc}$ for the O coverage below 1 ML.

Fig. 7 shows the E_{ads} of O atoms as the function of the O coverage on $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$. The E_{ads} of O atoms under different coverage was negative, indicating that the oxidation of $Surf_{CD}$ was spontaneous, especially for $Surf_{Vc}$. The E_{ads} increased with the O coverage, which indicates that the repulsive interaction of O atoms built up with the number of O atoms increasing.

Another effect of surface C-related point defects on the initial oxidation of 4H-SiC was found in the variation of bond length at the SiO_2/SiC interface. Fig. 8(a) and (b) show the distribution of Si-O and Si-C bonds extracted from the surface models displayed in Figs. 4(i), 5(i) and 6(i). The average Si-O bond lengths on the $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$ (1.67, 1.70 and 1.69 Å, respectively) were larger than that in clean 4H-SiC surface (1.63 Å) and bulk α-SiO_2 (1.62 Å) [27,55]. The average Si-C bond lengths on the $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$ (1.94, 1.92 and 1.94 Å, respectively) were larger than that in clean 4H-SiC surface and bulk 4H-SiC (1.89 Å) [55,56]. The more considerable average Si-O and Si-C bond lengths on the $Surf_{CD}$ may be attributed to the local lattice strain of the surfaces caused by C-related point defects, which is similar to the increase of Ga-N bond lengths at the GaN/GaAs interface caused by that the eight relaxed N atoms induced strain along *z* direction [57]. On the other hand, the increased repulsive interaction of O atoms with the number of surface atoms may also cause the lengthening of the Si-O and Si-C bonds.

From the above analysis and comparison, surface C-related point defects can be seen as a critical factor affecting the initial oxidation of 4H-SiC, which determine the adsorption sites of O atoms, the diffusion of O atoms to 4H-SiC and the interfacial lattice strain. The O atoms combined with the Si dangling bonds on the surface at low O coverage, preferentially adsorbed on the B site of two unoccupied Si atoms adjacent to the C-related defects. When the O coverage reached 5/9, 4/9 and 8/9 ML, all Si dangling bonded on the $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$ were saturated. Compared with the clean 4H-SiC surface (1/2 ML) [16], more O atoms were required for saturating Si dangling bonds on these $Surf_{CD}$ except $Surf_{Csi}$, which is directly related to the number of available Si

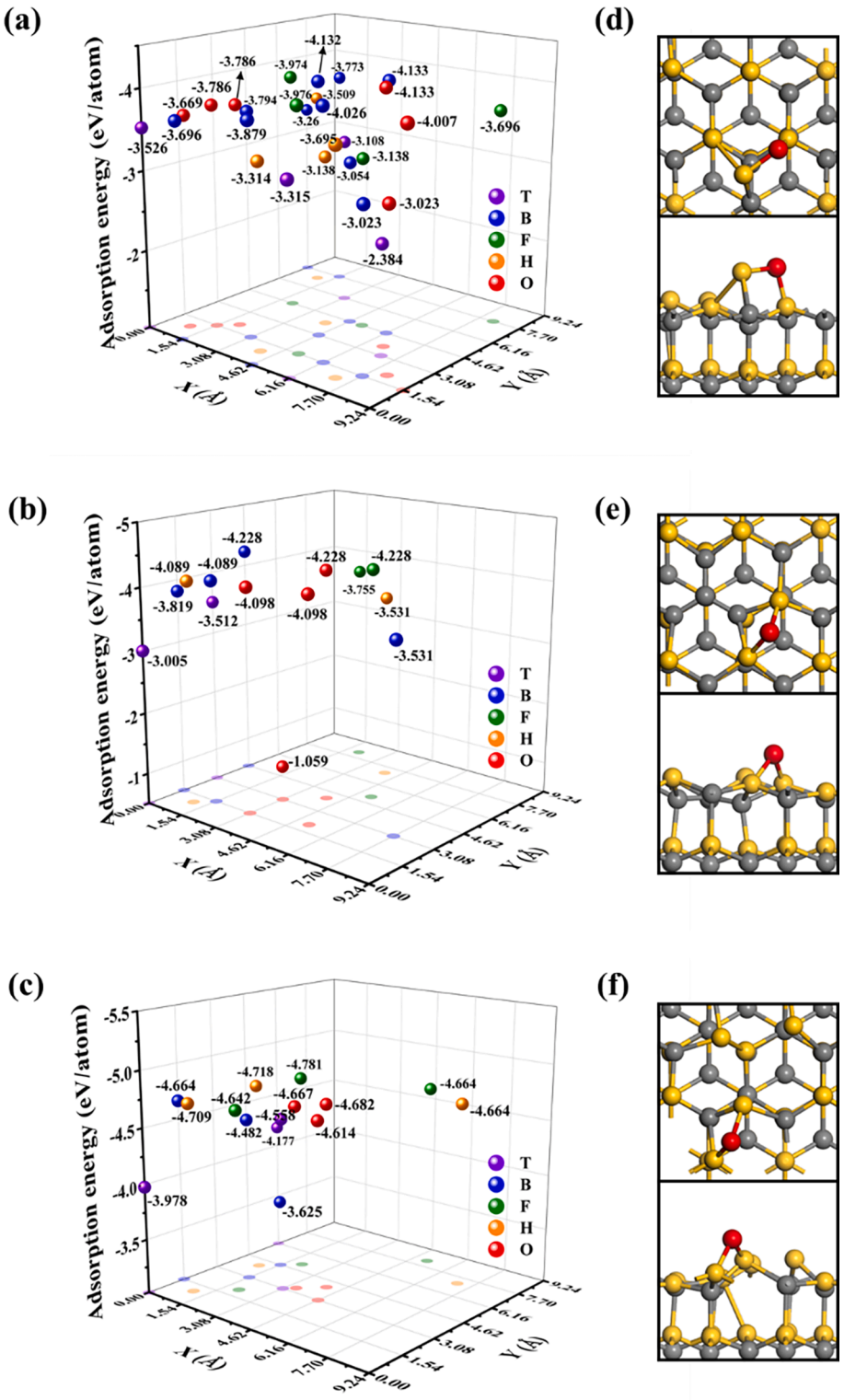


Fig. 3. The calculated adsorption energies of single O atom at different adsorption sites on the (a) $Surf_{Ci}$, (b) $Surf_{Csi}$, (c) $Surf_{Vc}$. The top and side views of most stable configurations for single O atom adsorption on the (d) $Surf_{Ci}$, (e) $Surf_{Csi}$, (f) $Surf_{Vc}$. The X and Y axes correspond to the X and Y directions shown in Fig. 2 **(a)** respectively, and the scales of the X and Y axes are calibrated according to the distance of 3.08 Å between two adjacent Si atoms on the surface. The circular patterns of different colors projected in the $z = 0$ plane correspond to the positions of each adsorption site in Fig. 2.

dangling bonds on the surface. The existence of surface defects also affected the diffusion of O atoms to the subsurface. For clean surface, the O atoms diffused to the subsurface when the O coverage reached 3/4 ML [16]. However, for $Surf_{CD}$, the minimum O coverage of O atom diffusion to subsurface was 7/9 ML. This difference in O diffusion caused by C-related point defects likely results in the difference of local oxidation rate and impact on the uniformity of oxide film thickness. In addition, the distribution of Si-O and Si-C bond lengths, as shown in Fig. 8, indicated that the C-related point defects could cause more significant lattice strain and distortion on the surface, which increases the possibility of forming more interface defects during oxidation.

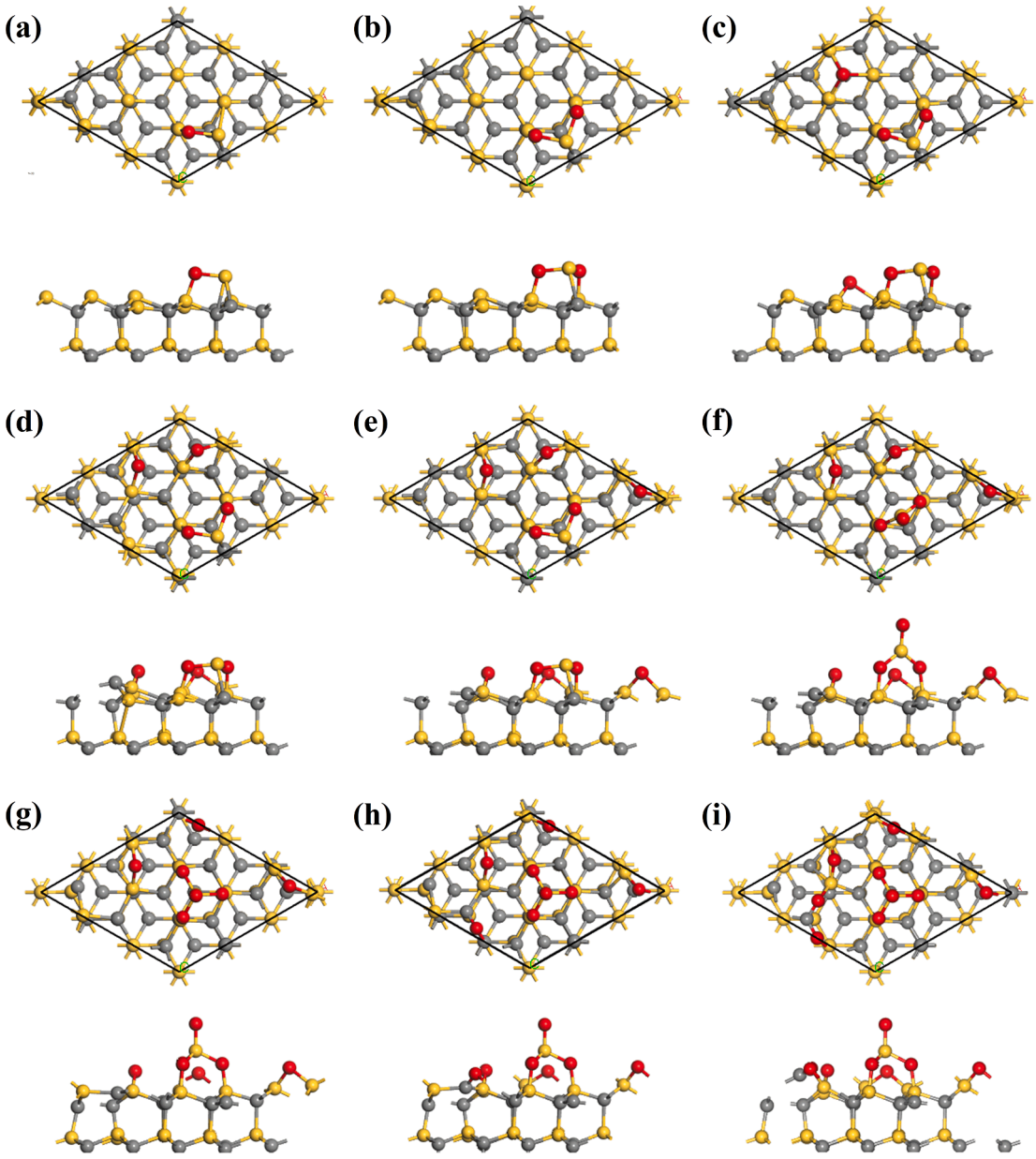


Fig. 4. Top and side views of the most stable configurations of O atoms adsorption on the $Surf_{Ci}$. (a)–(i) denote the configurations with O coverage of 1/9, 2/9, 1/3, 4/9, 5/9, 2/3, 7/9, 8/9 and 1 ML.

3.3. Thermodynamic equilibrium diagrams for initial oxidation of $Surf_{CD}$

This section investigated the variations of surface stability with temperature and pressure for O/$Surf_{Ci}$, O/$Surf_{Csi}$ and O/$Surf_{Vc}$. Firstly, the obtained relative surface free energies ($\Delta\gamma$) by Eq. (4) as a function of μ_O were depicted in Fig. 9. The range of μ_O to ensure the stability of SiO_2 was -9.16 eV $\leq \mu_O \leq -4.92$ eV, which is represented by the area between two vertical dashed lines in Fig. 9. The slopes of $\Delta\gamma \sim \mu_O$ curves increased with the of O coverage, consistent with the findings of clean surface [16]. The more negative the surface free energy is, the more stable the structure is [16,43]. For O-poor, the structure with low O coverage is more stable; on the contrary, the structure with high O coverage is more stable for O-rich. This is consistent with the variation of structural stability of different O coverage in GaN/HfO_2 system [58].

Fig. 9(a) shows the variation curves of $\Delta\gamma$ with μ_O for the O/$Surf_{Ci}$ structure under different O coverage. The $\Delta\gamma$ of $Surf_{Ci}$ is lowest for $\mu_O < -9.21$ eV, indicating the $Surf_{Ci}$ was the most stable. For -9.21 eV $< \mu_O < -9.16$ eV, the $\Delta\gamma$ of the structure with 4/9 ML O coverage is lowest, corresponding the structure with 4/9 ML O coverage became the most favorable. For $\mu_O > -9.16$ eV, the most stable structure transited from 4/9 ML to the bulk SiO_2, indicating that SiO_2 begins to nucleate when the O coverage on the $Surf_{Ci}$ reaches 4/9 ML or higher, which is consistent with the fact aforementioned in section **3.2** that the SiO_4 tetrahedron structure formed on the $Surf_{Ci}$ may be the initial nucleation precursor of thermally oxidized SiO_2. In addition, some possible intermediate configurations existed during the formation of bulk SiO_2, such as the structure with 5/9 ML, 7/9 ML, 8/9 ML and 1 ML, corresponding μ_O values of phase transition were -8.91, -8.65, -8.02 and -7.17 eV, respectively. These configurations were more stable than 4/9 ML but less stable than SiO_2.

Fig. 9(b) shows the variation curves of $\Delta\gamma$ with μ_O for the O/$Surf_{Csi}$ structure under different O coverage. The $Surf_{Csi}$ was most stable for $\mu_O < -9.21$ eV, while for -9.21 eV $< \mu_O < -9.16$ eV, the structure with 1/3 ML O coverage became the most stable. For $\mu_O > -9.16$ eV, the most stable structure was transformed to bulk SiO_2. Notably, the transition process from the $Surf_{Csi}$ to SiO_2 was almost aligned with that of the

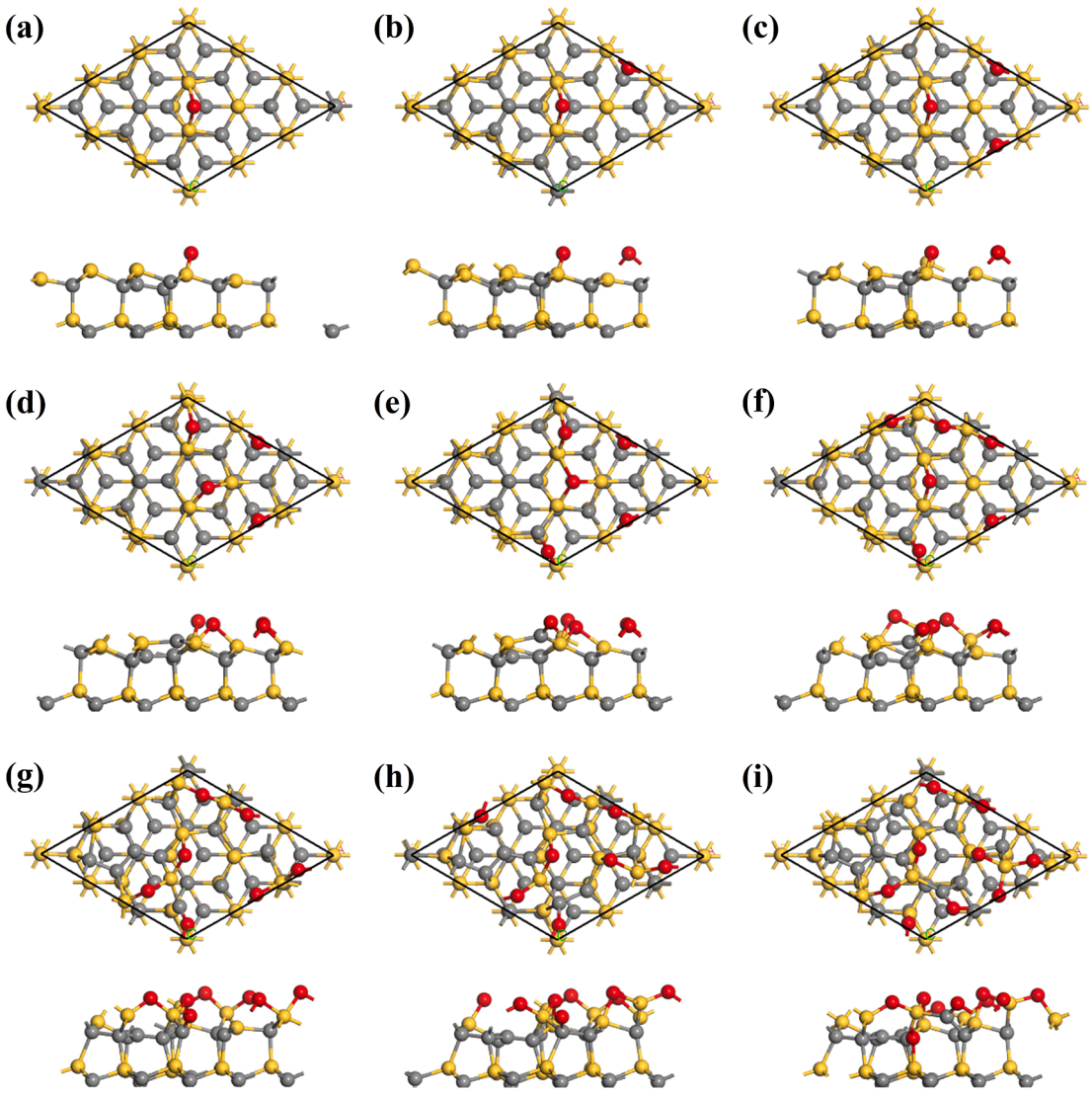


Fig. 5. Top and side views of the most stable configurations of O atoms adsorption on the $Surf_{Csi}$. (a)–(i) denote the configurations with O coverage of 1/9, 2/9, 1/3, 4/9, 5/9, 2/3, 7/9, 8/9 and 1 ML.

$Surf_{Ci}$, which may be associated with the fact that both surfaces contain C_{Si} defects. However, since no lattice reconstruction similar to the $Surf_{Ci}$ occurred on the $Surf_{Csi}$, a minority of intermediate transition structures (4/9 ML and 1 ML) exist during the formation of SiO_2 on the $Surf_{Csi}$, the corresponding μ_O values of phase transition were −8.87 and −8.55 eV.

Fig. 9(c) displays the variation curves of $\Delta\gamma$ with μ_O for the O/$Surf_{Vc}$ structure under different O coverage. When $\mu_O < -9.70$ eV, the $Surf_{Vc}$ was the most stable. Before the formation of SiO_2, $Surf_{Vc}$ had more kinds of stable structures of O adsorption than $Surf_{Ci}$ and $Surf_{Csi}$, such as 2/9 ML (-9.70 eV $< \mu_O < -9.34$ eV) and 5/9 ML (-9.34 eV $< \mu_O < -9.16$ eV), which is closely relevant to more Si dangling bonds on $Surf_{Vc}$. When μ_O was above −9.16 eV, the bulk SiO_2 was the most favorable. Moreover, some possible intermediate configurations, 2/3 ML, 7/9 ML, 8/9 ML and 1 ML, appeared during the formation of bulk SiO_2, corresponding μ_O values of phase transition were −8.73, −8.39, −8.17 and −7.75 eV, respectively.

For further investigation, the thermodynamic equilibrium diagrams of initial oxidation on $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$ were constructed in Fig. 10(a-c), which could clearly describe the variation of structures with temperature and pressure during oxidation. The region in the upper left corner was the most stable region of bulk SiO_2, and the boundary of phase transition to bulk SiO_2 was marked with the red dashed line. Fig. 10(a) and (b) show that there is only narrow band of temperature and pressure between the $Surf_{Ci}$ or $Surf_{Csi}$ and bulk SiO_2, which indicates that the O/$Surf_{Ci}$ and O/$Surf_{Csi}$ structures under low O coverage were unstable before the formation of SiO_2. For $Surf_{Vc}$, there were stable O adsorption structures of 2/9 ML and 5/9 ML on the surface before the formation of SiO_2 (Fig. 10(c)). We also found that some stable intermediate configurations existed on these $Surf_{CD}$. However, the surface configurations at actual oxidation conditions ($T \approx 1400$ K and $P \approx 1$ atm) were the most noteworthy in the device preparation process of the electronics industry [16,59,60]. The vertical gray dashed lines in Fig. 10 referred to $T = 1400$ K. The structures of 1 ML on these $Surf_{CD}$ were relatively stable at higher oxygen pressure, indicating that these intermediate configurations were most likely to be observed for $Surf_{CD}$. Meanwhile, all the configurations of 1 ML on $Surf_{CD}$ contained different types of defects, such as C-clusters, Si-O-C and Si dangling bonds, which likely affected the BTI and reliability of the 4H-SiC MOS devices. The evolution mechanisms and electronic properties of these defects are indispensable and will be conducted in Section 3.4.

3.4. Evolution and electronic properties of defects

The high density of SiO_2/4H-SiC interface defects formed by thermal

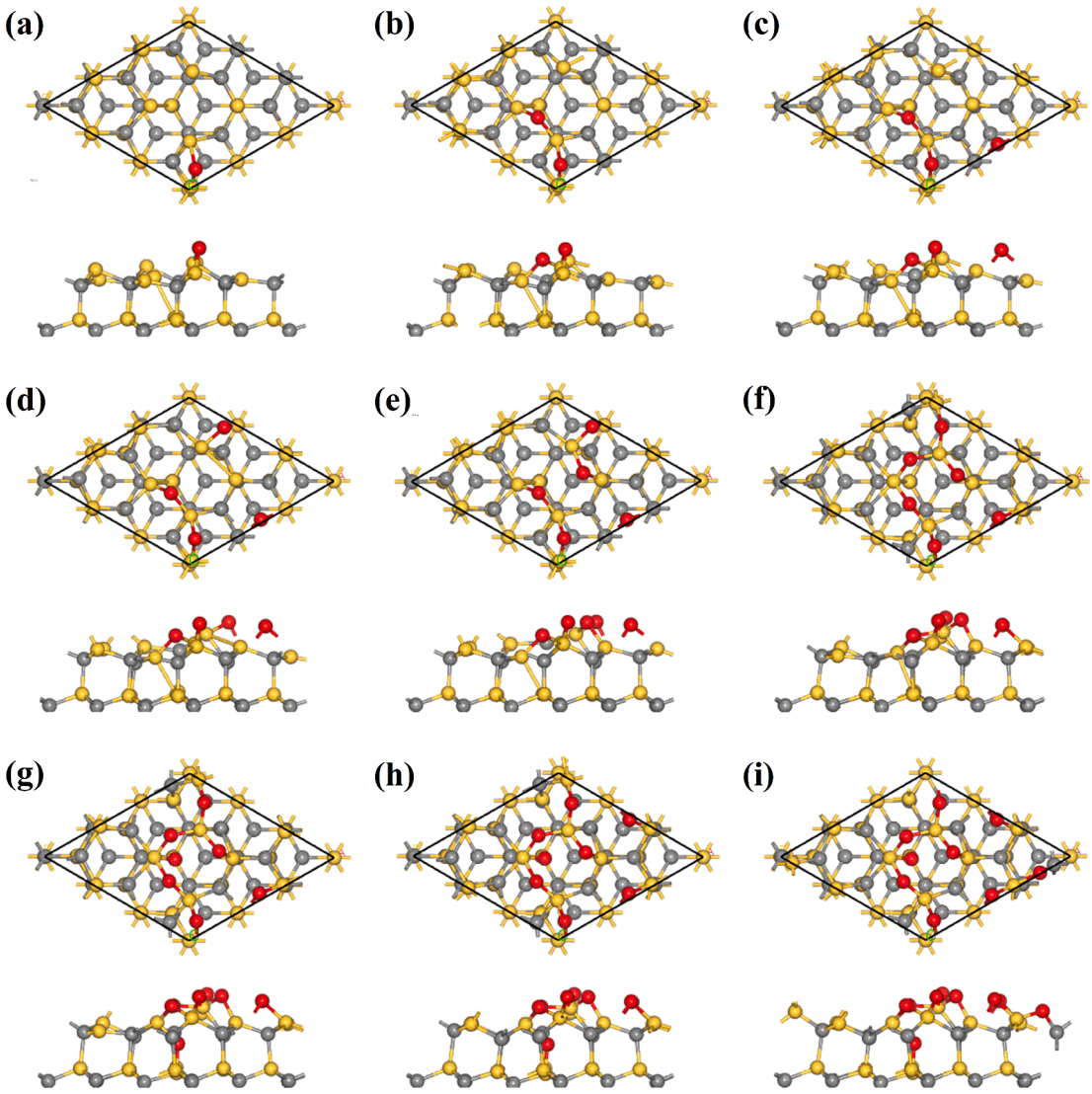


Fig. 6. Top and side views of the most stable configurations of O atoms adsorption on the $Surf_{Vc}$. (a)–(i) denote the configurations with O coverage of 1/9, 2/9, 1/3, 4/9, 5/9, 2/3, 7/9, 8/9 and 1 ML.

oxidation has always been one of the critical factors hindering the development of 4H-SiC MOS devices, and there is no exact and unified explanation for the origin of interface defects [14,59,61,62]. This section discussed the evolution of defects on the $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$ during the initial oxidation. The electronic properties of defects generated in the initial oxidation were further investigated.

As the initial oxidation process continued, the number of O atoms bound to the surface Si atoms increased. Some Si-C bonds on the surface and between bilayers were broken, thus forming the unsaturated C atoms, Si-O-C, and the Si dangling bond defects. Moreover, the unsaturated C atoms were further evolved into C-cluster defects as shown in Fig. 11, and the evolution mechanisms were mainly classified into two types:

Firstly, the unsaturated C atoms around initial C-related point defects tended to combine to form larger C-clusters. Fig. 11(a) shows that the Si-C bond connected to the interstitial C atom (C1) on the $Surf_{Ci}$ is broken to form C_4-cluster centered on C1. Moreover, the Si-C bond connected with C2 was destroyed to generate unsaturated C2 atom, which is prone to connect with the interstitial C1 atom to form a C=C bond, resulting in the transformation of C_4-cluster into C-(C=C)-C structure with mixed sp^2-sp^3 bonding. Fig. 11(b) shows the formation of a C_6-cluster on the $Surf_{Csi}$. The unsaturated C3 and C6 atoms combined with C4 and C5 around C_{Si} to form a C_6-cluster with sp^3 bonding.

The other was smaller C-clusters formed by the combination of unsaturated C atoms on the local regions without C-related point defects, which is analogous to the spontaneous formation of C–C defects due to the oxidation induced compressive strain [63]. Fig. 11(c) and (d) display the formation of C_2-clusters in the surface area of $Surf_{Csi}$ and $Surf_{Vc}$ away from C-related point defects, respectively.

To investigate the electronic properties of the defects, the TDOS and partial charge density of the 1 ML structures on these $Surf_{CD}$ were calculated. Fig. 12(a) shows the TDOS of O/$Surf_{Ci}$, O/$Surf_{Csi}$ and O/$Surf_{Vc}$ structures, the gap states in the 4H-SiC bandgap were observed for these three different surface structures. For $Surf_{Ci}$, the gap states were distributed in the range of 0.07–0.25 eV and 0.64–1.12 eV above the valence band maximum (VBM) and 0.15–0.45 eV and 0.60–0.88 eV below conduction band minimum (CBM). The partial charge densities of 0.07–0.25 eV above VBM and 0.15–0.45 eV below CBM were demonstrated in Fig. 12(b) and (e), which shows that the two gap states were mainly contributed by C_4-cluster defect. The gap state distributed in 0.64–1.12 eV above VBM was contributed by subsurface Si dangling bond defect and the surface oxygen dangling bond (Fig. 12(c)). And the gap state of 0.60–0.88 eV below CBM was contributed by the Si-O-C structure and C dangling bond defect (Fig. 12(d)). Likewise, the local

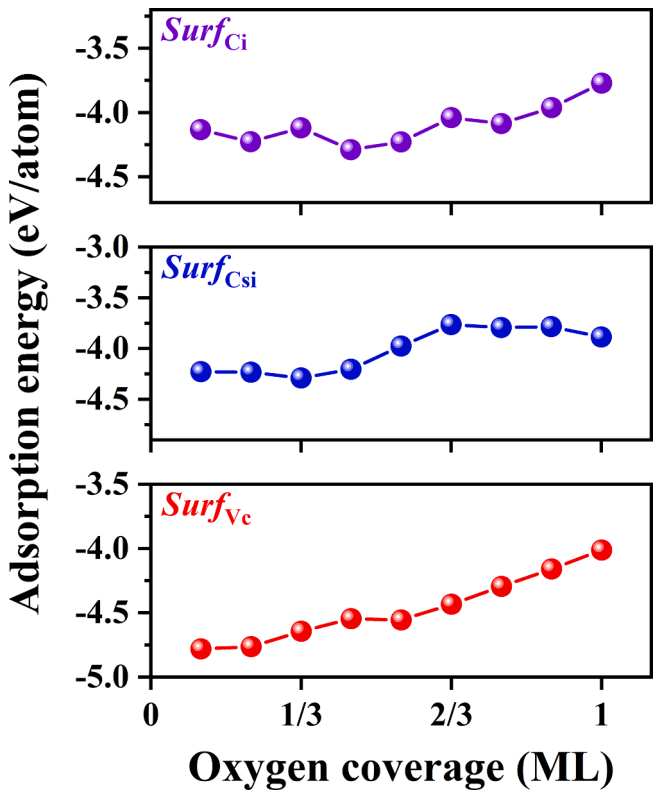


Fig. 7. The adsorption energies per O atom as a function of O coverage.

density of states (LDOS) of C_4-cluster, subsurface Si dangling bond and Si-O-C in O/*Surf*$_{Ci}$ structure has confirmed the distribution of states induced by these defects (Supplementary Fig. S7(a)). The gap states lay at 0–0.21 eV below CBM for *Surf*$_{Csi}$, corresponding to the partial charge density of C_6-cluster and C_2-cluster defects as shown in Fig. 12(f), and the same result was obtained from the LDOS of C_6-cluster and C_2-cluster in O/*Surf*$_{Csi}$ structure depicted in Supplementary Fig. S7(b). For *Surf*$_{Vc}$, the gap states were distributed in the range of 0–0.53 eV above the VBM, the partial charge density shown in Fig. 12(g) indicated that these gap states were mainly contributed by C_2-cluster defect, which is also verified by the result of LDOS of C_2-cluster in O/*Surf*$_{Vc}$ surface (Supplementary Fig. S7(c)). The gap states induced by the defects in Fig. 12 (b), (c) and (g) were closer to VBM, which are more likely to capture carriers from the valence band. The trapped carriers were constrained at the interface, degrading the performance of p-channel MOSFETs devices. On the contrary, the gap states produced by the defects in Fig. 12 (d), (e) and (f) were closer to CBM, unfavorable for the n-channel SiC MOSFETs.

4. Conclusion

The DFT calculations in this work have been performed to provide deep insights into the initial oxidation on the 4H-SiC (0001) surface with C-related point defects. We have found that the most stable adsorption sites of single O atom were bridge sites between the Si atoms connected to the C-related point defects on these *Surf*$_{CD}$. The adsorption energy of single O atom on *Surf*$_{Vc}$ was the lowest among these *Surf*$_{CD}$ and lower than that on clean surface. As the oxidation process continued, the O atoms preferentially combined with the Si dangling bonds on the surface at low O coverage, and more O atoms were required for saturating Si dangling bonds on these *Surf*$_{CD}$ except *Surf*$_{Csi}$. The diffusion rates of O atoms on all *Surf*$_{CD}$ were slower than that on the clean surface. C-related point defects could also cause more significant lattice strain and distortion on the surface, which increases the possibility of forming more interface defects during oxidation. Furthermore, the thermodynamic calculations showed that the structures under low O coverage on the *Surf*$_{Ci}$ and *Surf*$_{Csi}$ were unstable before the formation of SiO_2. Some intermediate structures with defects were observed during the formation of SiO_2 on these *Surf*$_{CD}$. The newly generated interface Si dangling bonds, Si-O-C structure and C-clusters, introducing charge states near CBM and VBM of 4H-SiC, which may further affect the voltage stability of 4H-SiC MOS devices.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Shengsheng Wei: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing. **Zhipeng Yin:**

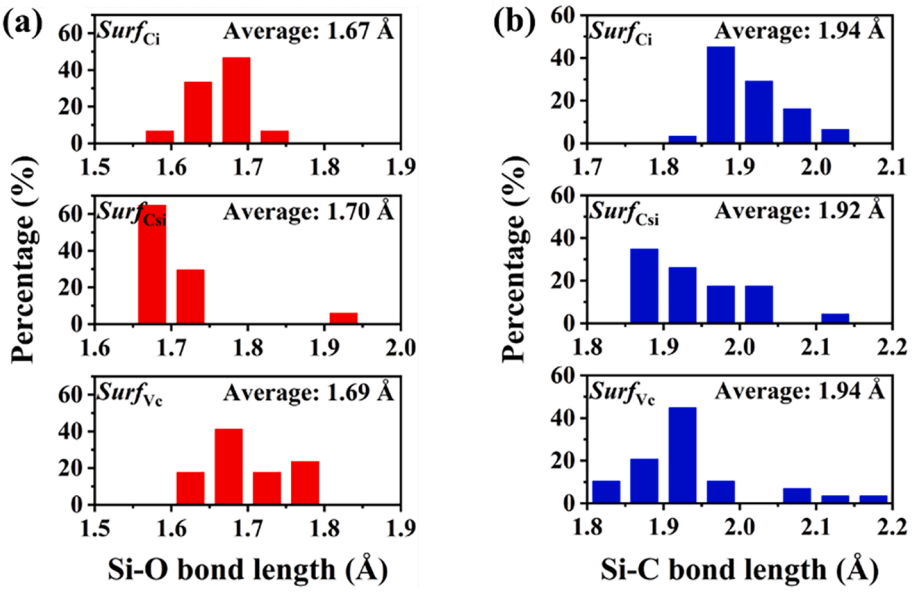


Fig. 8. The distribution of (a) Si-O bond length and (b) Si-C bond length for *Surf*$_{Ci}$, *Surf*$_{Csi}$ and *Surf*$_{Vc}$ with O coverage of 1 ML.

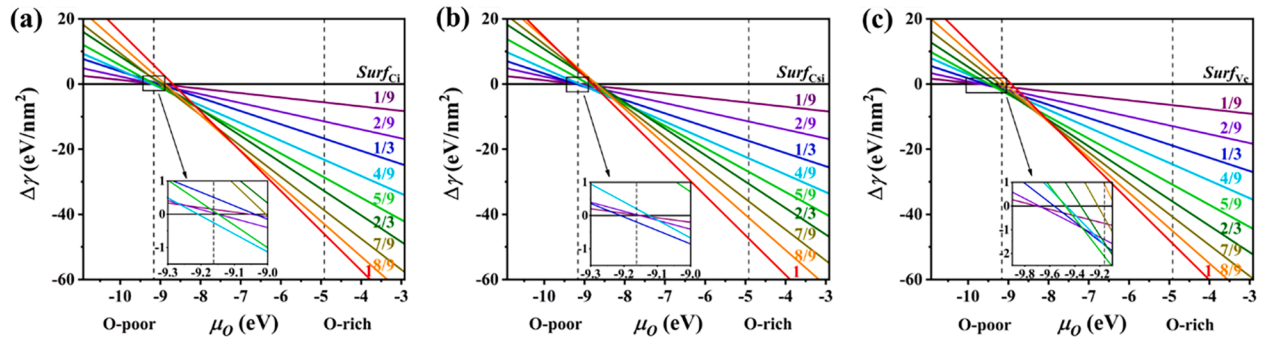


Fig. 9. The relative surface free energies for the O adsorption on the (a) $Surf_{Ci}$, (b) $Surf_{Csi}$ and (c) $Surf_{Vc}$ under different O coverage versus the O chemical potential. Here horizontal black line in (a), (b) and (c) correspond to the $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$ structure respectively. The vertical dashed lines represent the boundaries of O chemical potential to ensure the stability of bulk SiO_2, the line of $\mu_O = -4.92$ eV is the upper boundary, the line of $\mu_O = -9.16$ eV is the lower boundary. The inset figures represent the enlarged views of the regions marked by box.

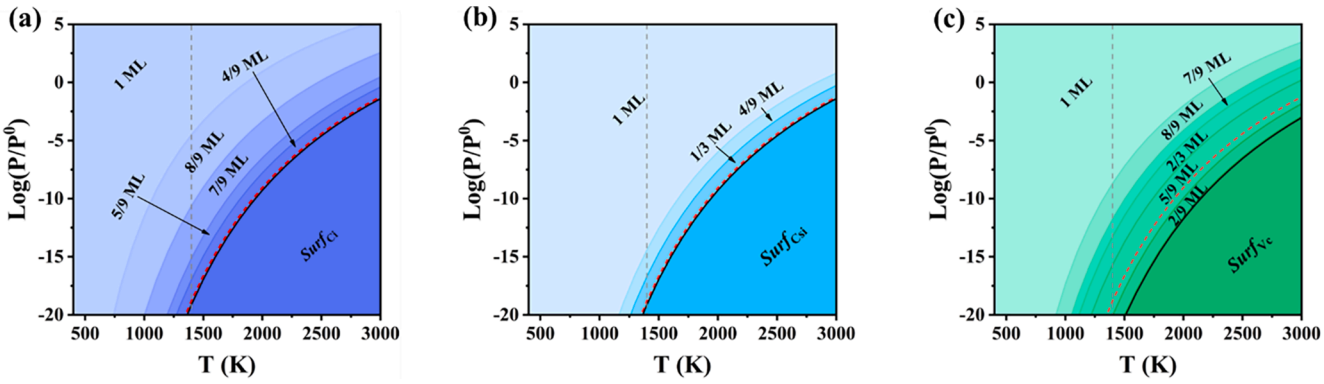


Fig. 10. The thermodynamic equilibrium diagram of initial oxidation on (a) $Surf_{Ci}$, (b) $Surf_{Csi}$ and (c) $Surf_{Vc}$. The calculated phase boundary of the transition from O/SiC to SiO_2 is marked with red dashed line. The vertical gray dashed lines refer to $T = 1400$ K. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

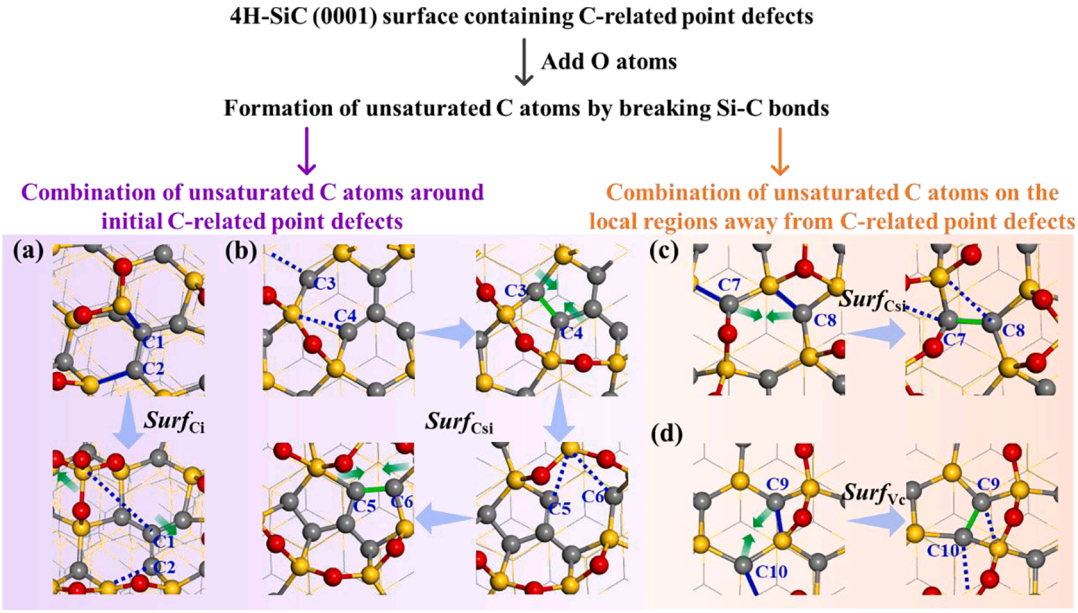


Fig. 11. The formation mechanism and process of C-cluster defects on the $Surf_{Ci}$, $Surf_{Csi}$ and $Surf_{Vc}$ at the initial oxidation. (a) The formation of C_4-cluster on the $Surf_{Ci}$. (b) The formation of C_6-cluster on the $Surf_{Csi}$. (c-d) the formation of C_2-cluster on the $Surf_{Csi}$ and $Surf_{Vc}$. The solid blue line and dotted blue line represent the connection and breaking of Si-C bond respectively. The solid green line represents the C–C bond. The green arrow refers to the moving direction of the atom. C1–C10 denote the C atoms in the C-clusters. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

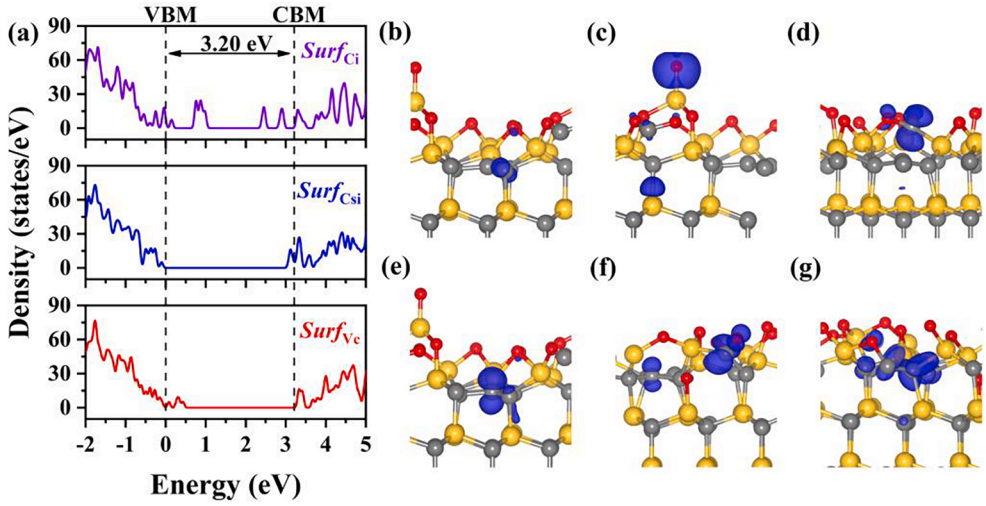


Fig. 12. (a) The total density of states (TDOS) of O/$Surf_{Ci}$, O/$Surf_{Csi}$ and O/$Surf_{Vc}$ structures. The VBM and CBM are marked with the left and right vertical dashed lines, and the VBM is aligned at 0 eV. (b–e) Partial charge density in 0.07–0.25 eV above VBM, 0.64–1.12 eV above VBM, 0.60–0.88 eV below CBM and 0.15–0.45 eV below CBM for O/$Surf_{Ci}$ structure. (f) Partial charge density in 0–0.21 eV below CBM for O/$Surf_{Csi}$ structure. (g) Partial charge density in 0–0.53 eV above VBM for O/$Surf_{Vc}$ structure. The isosurface is set to 0.015 e/Bohr^3.

Conceptualization, Methodology, Resources. **Jiao Bai:** Methodology, Resources. **Weiwei Xie:** Methodology, Resources. **Fuwen Qin:** Resources, Writing – review & editing. **Yan Su:** Software. **Dejun Wang:** Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (No. 61874017).

APPENDIX: LIST OF PARAMETERS[1]

Parameters	Description
E_{ads} (O)	The adsorption energy per O atom on the $Surf_{CD}$
$E_{surf/O}$	The total energy of 4H-SiC surface model with O atoms adsorbed
E_{surf}	The total energy of 4H-SiC surface model without O atoms adsorbed
N_O	The number of O atoms
E_{tot} (O_2)	The energy of a O_2 molecule
T	Temperature
P	Pressure
$\gamma_{surf/O}$	The surface free energy of $Surf_{CD}$ with different O coverages
S	The surface area of the surface model
$G_{surf/O}$	The Gibbs free energy of $Surf_{CD}$ with different O coverages
N_{Si}	The number of Si atoms
N_C	The number of C atoms
μ_{Si}	The chemical potential of Si atom
μ_C	The chemical potential of C atom
μ_O	The chemical potential of O atom
γ_{surf}	The surface free energy of $Surf_{CD}$ without O coverage
$\Delta\gamma$	The relative surface free energy
ΔN_C	The difference in the number of C atoms on the O/4H-SiC surface relative to 4H-SiC surface
ΔN_{Si}	The difference in the number of Si atoms on the O/4H-SiC surface relative to 4H-SiC surface
ΔN_O	The difference in the number of O atoms on the O/4H-SiC surface relative to 4H-SiC surface
P^0	Standard atmospheric pressure (1 atm)
μ_{O2} (T, P^0)	The standard chemical potential of O_2
k_B	Boltzmann constant
μ_O^{MAX}	The maximum value for chemical potential of O
μ_O^{MIN}	The minimum value for chemical potential of O
E_{tot} (SiO_2)	The energies of a bulk SiO_2 unit
E_{tot} (Si)	The energies of a bulk Si unit

[1] The parameters are arranged in the order in which they appear in the equations.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apsusc.2022.156161.

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